R E A C T：语言模型中推理与行动的协同

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摘要

虽然大型语言模型（LLMs）在语言理解和交互式决策任务中展现了令人瞩目的性能，但其推理能力（例如思维链提示）和行动能力（例如行动规划生成）此前主要作为独立主题进行研究。在本文中，我们探索了以交错方式利用LLMs同时生成推理轨迹和任务特定行动，使两者之间产生更强的协同效应：推理轨迹帮助模型推导、追踪和更新行动计划并处理异常，而行动则允许模型与外部知识库或环境等来源进行交互并收集额外信息。我们将这一方法命名为ReAct，并将其应用于多种语言和决策任务，证明了其相较于最先进基线的有效性，同时提升了人类可解释性和可信度。具体而言，在问答（HotpotQA）和事实验证（Fever）任务中，ReAct通过与简单的Wikipedia API交互，克服了思维链推理中常见的幻觉和错误传播问题，并生成了比不带推理轨迹的基线更具可解释性的人类式任务解决轨迹。此外，在两个交互式决策基准（ALFWorld和WebShop）中，ReAct仅通过一两个上下文示例进行提示，其绝对成功率分别比模仿学习和强化学习方法高出34%和10%。

1 引言

人类智能的一个独特特征是能够将面向任务的行为与语言推理（或内心独白，Alderson-Day & Fernyhough, 2015）无缝结合，这在理论上被认为在人类认知中扮演重要角色，有助于实现自我调节或策略规划（Vygotsky, 1987; Luria, 1965; Fernyhough, 2010）以及维持工作记忆（Baddeley, 1992）。以在厨房烹饪菜肴为例。在两个具体动作之间，我们可能会用语言进行推理，以追踪进度（“现在所有东西都切好了，我应该烧热水”），处理异常情况或根据实际情况调整计划（“我没有盐，那就用酱油和胡椒代替”），以及在需要外部信息时意识到这一点（“怎么揉面团？让我上网查一下”）。我们也可能采取行动（打开食谱书阅读菜谱、打开冰箱、检查食材）来支持推理并回答问题（“我现在能做什么菜？”）。这种“行动”与“推理”之间的紧密协同使得人类能够快速学习新任务，并在即使面对前所未见的情况或信息不确定性时，也能做出稳健的决策或推理。

最近的研究结果暗示了在自主系统中将语言推理与交互式决策相结合的可能性。一方面，经过适当提示的大型语言模型（LLMs）展示了执行多步推理轨迹以推导出的涌现能力。

[*]在Goo期间工作 gle internship. Projet page with code: https://react-lm.github.io/. 1

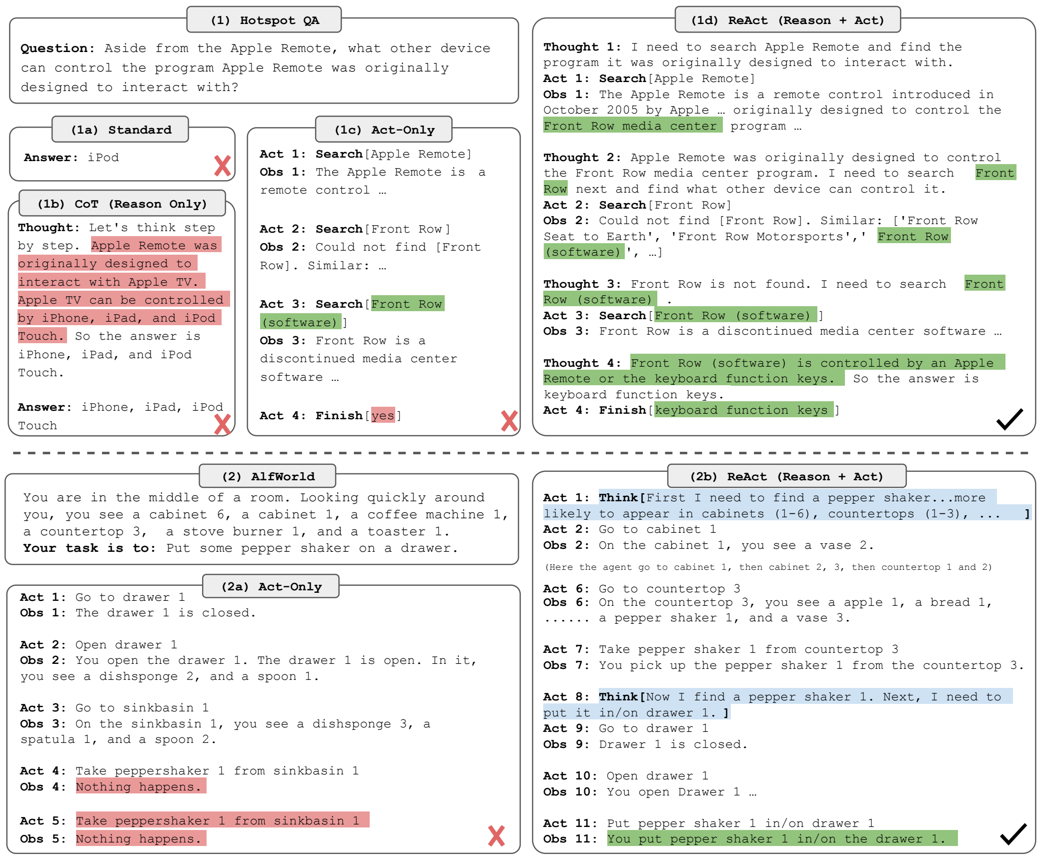

图1：（1）四种提示方法的对比，(a) 标准提示、(b) 思维链（CoT，仅推理）、(c) 仅行动提示和 (d) 推理+行动提示（Reason+Act），解决一个HotpotQA（Yang等人，2018）问题。（2）(a) 仅行动提示与 (b) 推理+行动提示解决一个AlfWorld（Shridhar等人，2020b）游戏的对比。在两个领域中，我们均省略了提示中的上下文示例，仅展示由模型（行动、思考）和环境（观察）生成的解决轨迹。

来自算术、常识和符号推理任务的答案（Wei et al., 2022）。然而，这种“思维链”推理是一个静态的黑箱，因为模型使用其内部表征生成思考，并未基于外部世界，这限制了其反应性推理或更新知识的能力，可能导致事实幻觉和推理过程中的错误传播等问题（图1（1b））。另一方面，近期研究探索了使用预训练语言模型在交互环境中进行规划与行动（Ahn et al., 2022; Nakano et al., 2021; Yao et al., 2020; Huang et al., 2022a），重点是通过语言先验预测行动。这些方法通常将多模态观察转化为文本，使用语言模型生成领域特定的行动或计划，然后通过控制器选择或执行它们。然而，它们并未利用语言模型对高层目标进行抽象推理，或维护工作记忆以支持行动，除Huang et al. (2022b)外，他们执行了有限形式的言语推理以重申关于当前状态的空间事实。除了这种与少量方块交互的简单具身任务外，尚未有研究探讨如何以协同方式将推理与行动结合以解决通用任务，以及这种结合是否能相比单独推理或行动带来系统性优势。

在本工作中，我们提出ReAct，一种将语言模型的推理与行动相结合以解决多样化语言推理和决策任务的通用范式（图1）。ReAct促使LLMs以交错方式生成与任务相关的语言推理轨迹和行动，使模型能够进行动态推理，以创建、维护和调整高层行动计划（推理以行动），同时与外部环境（如维基百科）交互，将额外信息融入推理中（行动以推理）。

我们对ReAct和四种不同基准的最新基线进行了实证评估：问答（HotPotQA，Yang等人，2018）、事实验证（Fever，Thorne等人，2018）、文本游戏（ALFWorld，Shridhar等人，2020b）和网页导航（WebShop，Yao等人，2022）。在HotPotQA和Fever中，通过访问模型可以交互的维基百科API，ReAct优于普通的动作生成模型，同时与思维链推理（CoT）（Wei等人，2022）具有竞争性。总体最佳方法是ReAct和CoT的结合，允许在推理过程中同时使用内部知识和外部获取的信息。在ALFWorld和WebShop上，两次甚至一次ReAct提示就能超越使用$10^3 \sim 10^5$个任务实例训练的模仿或强化学习方法，成功率分别绝对提升34%和10%。我们还通过展示在仅使用动作的受控基线上的一致优势，证明了稀疏、多用途推理在决策中的重要性。除了广泛的适用性和性能提升外，推理与行动的结合还有助于所有领域的模型可解释性、可信度和可诊断性，因为人类可以轻松区分来自模型内部知识与外部环境的信息，并检查推理轨迹以理解模型动作的决策依据。

总之，我们的主要贡献如下：（1）我们引入了ReAct，一种新颖的基于提示的范式，用于协同语言模型中的推理与行动，以解决通用任务；（2）我们在多个基准上进行了大量实验，展示了ReAct在少样本学习设置下相较于先前单独进行推理或行动生成的方法的优势；（3）我们进行了系统性的消融实验和分析，以理解行动在推理任务中的重要性，以及推理在交互任务中的重要性；（4）我们分析了ReAct在提示设置下的局限性（即推理和行动行为的支持有限），并进行了初步微调实验，展示了ReAct通过额外训练数据改进的潜力。将ReAct扩展到更多任务上进行训练和操作，并与强化学习等互补范式相结合，可以进一步释放大型语言模型的潜力。

2 反应：协同推理 + 行动 G

考虑一个智能体与环境交互以解决问题的通用设置。在时间步t，智能体从环境接收观察$o_t \in \mathcal{O}$，并根据某个策略$a_t \in \mathcal{A}$采取行动$\pi(a_t|c_t)$，其中$c_t = (o_1, a_1, \cdots, o_{t-1}, a_{t-1}, o_t)$是相对于智能体的*context*。当映射$c_t \mapsto a_t$高度隐含且需要大量计算时，学习策略具有挑战性。例如，图1(1c)中所示的智能体无法生成正确的最终行动（Act 4）来完成任务，因为它需要对轨迹上下文（Question, Act 1-3, Obs 1-3）进行复杂推理。类似地，图1(2a)中所示的智能体无法从上下文中理解水槽1不包含胡椒瓶1，因此不断产生幻觉行动。

ReAct 的理念十分简洁：我们将智能体的动作空间扩展至$\hat{\mathcal{A}} = \mathcal{A} \cup \mathcal{L}$，其中 $\mathcal{L}$ 是语言空间。语言空间中的动作 $\hat{a}_t \in \mathcal{L}$（我们称之为 *thought* 或 *reasoning trace*）不会影响外部环境，因此不会产生观察反馈。相反，思维 $\hat{a}_t$ 旨在通过对当前上下文 c_t 进行推理来整合有用信息，并更新上下文 $c_{t+1} = (c_t, \hat{a}_t)$ 以支持后续推理或行动。如图 1 所示，存在多种类型的有用思维，例如：分解任务目标并制定行动计划（2b，行动 1；1d，思维 1），注入与任务解决相关的常识性知识（2b，行动 1），从观察中提取重要部分（1d，思维 2，4），跟踪进度并转换行动计划（2b，行动 8），处理异常情况并调整行动计划（1d，思维 3），等等。

然而，由于语言空间$\mathcal{L}$是无限的，在这种增强的动作空间中进行学习是困难的，并且需要强大的语言先验。在本文中，我们主要关注一种设置：使用冻结的大型语言模型 PaLM-540B（Chowdhery 等人，2022）[1]，通过少量上下文示例提示该模型，以生成特定领域的动作和自由形式的语言思考，用于任务求解（图 1 (1d), (2b)）。每个上下文示例是一个人类轨迹，包含动作、思考和环境观测，用于解决一个任务实例（参见附录 C）。对于推理至关重要的任务（图 1(1)），我们交替生成思考与动作，使得任务求解轨迹由多个思考-动作-观测步骤组成。相比之下，对于可能涉及大量动作的决策任务（图 1(2)），思考只需

1我们展示一些GPT-3 (Brown et al., 2020) results in Appendix A.1, which outperfor毫秒 PaLM-540B.

在轨迹最相关的位置稀疏出现，因此我们让语言模型自行决定思想与动作的异步发生。

由于决策和推理能力已集成到大型语言模型中，ReAct具备几项独特特性：A) 直观且易设计：设计ReAct提示模板非常直接，人类标注者只需在行动基础上用语言记录其思考过程。本文未使用任何临时格式选择、思维设计或示例筛选。我们在第3节和第4节详细说明了每个任务的提示设计。B) 通用且灵活：由于灵活的思维空间及思维-行动出现形式，ReAct适用于具有不同行动空间和推理需求的多样化任务，包括但不限于问答、事实验证、文字游戏和网页导航。C) 高性能且稳健：ReAct仅依赖一到六个上下文示例进行学习，即能对新任务实例展现出强大的泛化能力，在不同领域持续优于仅包含推理或仅包含行动的基线方法。我们还在第3节展示了启用微调时的额外优势，并在第4节说明了ReAct性能对提示选择具有稳健性。

D) 人类对齐且可控：ReAct 承诺提供一个可解释的序列决策与推理过程，使人类能够轻松检查推理及事实的正确性。此外，人类还可以通过思维编辑实时控制或纠正智能体行为，如图4中图5所示。

3 知识密集型推理任务

我们从知识密集型推理任务开始，例如多跳问答和事实验证。如图1(1d)所示，通过与维基百科API交互，ReAct能够检索信息以支持推理，同时也能利用推理来定位下一步需要检索的内容，展示了推理与行动的协同效应。

3.1 设置

领域　我们考虑两个具有挑战性的知识检索与推理数据集：1）HotPotQA（Yang 等人，2018），一个需要基于两个或更多维基百科段落进行推理的多跳问答基准；（2）FEVER（Thorne 等人，2018），一个事实验证基准，其中每个声明根据是否存在维基百科段落来验证该声明，被标注为 SUPPORTS（支持）、REFUTES（反驳）或 NOT ENOUGH INFO（信息不足）。在本工作中，我们对这两个任务均采用仅问题设置，即模型仅接收问题/声明作为输入，无法访问支持段落，必须依赖其内部知识或通过与外部环境交互来检索知识以支持推理。

动作空间 我们设计了一个简单的Wikipedia Web API，支持三种类型的动作以实现交互式信息检索：(1) search[entity]，如果存在对应实体的Wiki页面，则返回该页面的前5个句子，否则从Wikipedia搜索引擎返回前5个相似实体建议；(2) lookup[string]，返回页面中包含字符串的下一个句子，模拟浏览器中的Ctrl+F功能；(3) finish[answer]，以答案结束当前任务。我们注意到，这个动作空间大多只能基于精确的段落名称检索段落的一小部分，其能力显著弱于当前最先进的词汇或神经检索器。其目的是模拟人类与Wikipedia的交互方式，并迫使模型通过语言中的显式推理进行检索。

3.2 方法

ReAct 提示法 对于 HotpotQA 和 Fever，我们从训练集中随机选取 6 个和 3 个案例[2]，并手动编写 ReAct 格式的轨迹，用作提示中的少样本示例。与图 1(d) 类似，每条轨迹由多个思考-行动-观察步骤（即密集思考）组成，其中自由形式的思考用于多种目的。具体来说，我们使用了多种思考的组合，包括分解问题（“我需要搜索 x，找到 y，然后找到 z”）、从维基百科观察中提取信息（“x 始于 1844 年”、“该段落未提及 x”）、进行常识推理（“x 不是 y，因此 z 必须是……”）或算术推理（“1844 < 1989”）、引导

[2]We find more examples do not improve performance.

Prompt Method[a]	**HotpotQA** (EM)	**Fever** (Acc)
Standard	28.7	57.1
CoT (Wei et al., 2022)	29.4	56.3
CoT-SC (Wang et al., 2022a)	33.4	60.4
Act	25.7	58.9
ReAct	27.4	60.9
CoT-SC → ReAct	34.2	**64.6**
ReAct→ CoT-SC	**35.1**	62.0
Supervised SoTA[b]	67.5	89.5

表1：PaLM-540B在HotpotQA和Fever上的提示结果。

[a]HotpotQA EM在Wang et al. (2022b)中的Standard、CoT、CoT-SC结果分别为27.1、28.9、33.8。[b](Zhu et al., 2021; Lewis et al., 2020)

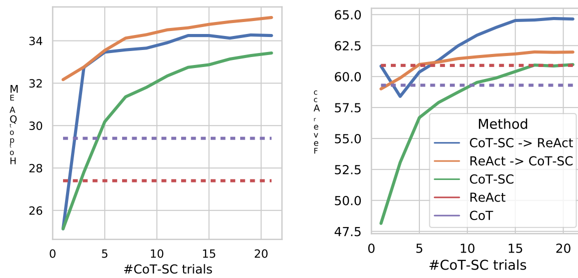

图2：PaLM-540B在使用不同数量的CoT-SC样本时的提示结果。

搜索重构（“也许我可以搜索/查找 x 替代”），并综合最终答案（“……所以答案是 x”）。更多细节见附录 C。

基线方法 我们系统地消融ReAct轨迹，为多个基线方法构建提示（格式如图1(1a-1c)所示）：(a) 标准提示（Standard），去除ReAct轨迹中的所有思考、动作和观察。(b) 思维链提示（CoT）（Wei等人，2022），去除动作和观察，作为纯推理基线。我们还构建了自一致性基线（CoT-SC）（Wang等人，2022a;b），通过在推理过程中使用解码温度0.7采样21条CoT轨迹，并采用多数答案，这被发现能持续提升CoT的性能。(c) 仅动作提示（Act），去除ReAct轨迹中的思考，大致类似于WebGPT（Nakano等人，2021）与互联网交互来回答问题的方式，尽管它操作在不同的任务和动作空间上，并使用模仿学习和强化学习而非提示。

结合内部与外部知识。正如将在第3.3节中详述的那样，我们观察到ReAct所展示的问题解决过程更加基于事实和扎实，而CoT在构建推理结构方面更为准确，但容易产生幻觉事实或想法。因此，我们提出结合ReAct和CoT-SC，并让模型根据以下启发式规则决定何时切换到另一种方法：A) ReAct → CoT-SC: 当ReAct在给定步数内未能返回答案时，回退到CoT-SC。我们对HotpotQA和FEVER分别设置7步和5步，因为我们发现更多步数不会改善ReAct的性能[3]。B) CoT-SC → ReAct: 当n个CoT-SC样本中的多数答案出现次数少于$n/2$次（即内部知识可能无法可靠地支持任务）时，回退到ReAct。

微调　由于手动大规模标注推理轨迹和操作存在挑战，我们采用了一种类似Zelikman等人（2022）的自举方法，使用由ReAct（也用于其他基线方法）生成的3000条带有正确答案的轨迹来微调较小的语言模型（PaLM-8/62B），使其能够根据输入的问题/声明解码出完整的轨迹（包括所有思考、操作和观察）。更多细节见附录B.1。

3.3 结果与观察

ReAct始终优于Act。表1展示了使用PaLM-540B作为基础模型并采用不同提示方法在HotpotQA和Fever上的结果。我们注意到，ReAct在两个任务上都优于Act，这证明了推理对指导行动的价值，尤其是在综合最终答案方面，如图1（1c-d）所示。微调结果3也证实了推理轨迹对更有依据的行动的益处。

[3]Of all trajectories with correct final answers, those with 7 steps on HotpotQA and 5 steps on FEVER only take up 0.84% and 1.33% respectively.

	Type	Definition	ReAct	CoT
Success	True positive	Correct reasoning trace and facts	94%	86%
	False positive	Hallucinated reasoning trace or facts	6%	14%
Failure	Reasoning error	Wrong reasoning trace (including failing to recover from repetitive steps)	47%	16%
	Search result error	Search return empty or does not contain useful information	23%	-
	Hallucination	Hallucinated reasoning trace or facts	0%	56%
	Label ambiguity	Right prediction but did not match the label precisely	29%	28%

表2：ReAct和CoT在HotpotQA上的成功与失败模式类型，以及它们在人类研究的随机选取样本中的百分比。

另一方面，ReAct 在 Fever 上优于 CoT（60.9 vs. 56.3），而在 HotpotQA 上略逊于 CoT（27.4 vs. 29.4）。Fever 中的 SUPPORTS/REFUTES 断言可能仅存在细微差异（参见附录 D.1），因此采取行动检索准确且最新的知识至关重要。为了更深入理解 ReAct 与 CoT 在 HotpotQA 上的行为差异，我们分别从 ReAct 和 CoT 中随机抽取了 50 条正确与错误答案（以 EM 为评判标准）的轨迹（共计 200 个样本），并手动标注了它们的成功与失败模式，如表 2 所示。关键观察结果如下：

A) 幻觉是CoT的一个严重问题，导致其成功模式下的假阳性率远高于ReAct（14%对6%），并构成了其主要失败模式（56%）。相比之下，ReAct的问题解决轨迹更为扎实、以事实为驱动且可信赖，得益于对外部知识库的访问。

B) 虽然交错推理、行动和观察步骤提高了ReAct的根基性和可信度，但这种结构约束也降低了其在制定推理步骤时的灵活性，导致推理错误率高于CoT。我们注意到，ReAct有一种常见的错误模式，即模型重复生成之前的想法和行动，我们将其归类为“推理错误”的一部分，因为模型无法推理出正确的下一步行动并跳出循环[4]。

C) 对于ReAct而言，通过搜索成功获取信息性知识至关重要。非信息性搜索占错误案例的23%，它会破坏模型的推理过程，使其难以恢复并重新组织思路。这或许是事实性与灵活性之间一种预期的权衡，也促使我们提出了结合两种方法的策略。

我们在附录E.1中提供了每个成功和失败模式的示例。我们还发现一些HotpotQA问题可能包含过时的答案标签，例如参见图4。

ReAct + CoT-SC 在提示大型语言模型时表现最佳 同样如表1所示，在 HotpotQA 和 Fever 上最优的提示方法分别是 ReAct → CoT-SC 和 CoT-SC → ReAct。此外，图2展示了不同方法随 CoT-SC 样本数量变化的表现。虽然两种 ReAct + CoT-SC 方法各自在一个任务上有优势，但它们在不同样本数量下均显著且一致地优于 CoT-SC，仅用3-5个样本就达到了 CoT-SC 使用21个样本的性能。这些结果表明，在推理任务中合理结合模型内部知识与外部知识具有重要价值。

ReAct 在微调中表现最佳。图3展示了在HotpotQA上四种方法（Standard、CoT、Act、ReAct）的提示/微调扩展效果。使用PaLM-8/62B时，由于难以从上下文示例中同时学习推理和行动，提示式ReAct在四种方法中表现最差。然而，仅用3,000个样本进行微调后，ReAct便成为四种方法中表现最佳的方法，其中PaLM-8B微调后的ReAct超越了所有PaLM-62B的提示方法，而PaLM-62B微调后的ReAct则超越了所有540B的提示方法。相比之下，对于PaLM-8/62B，微调Standard或CoT的效果明显差于微调ReAct或Act，因为前者本质上是教模型记忆（可能虚构的）知识事实，而后者则是教模型如何（推理并）行动以从维基百科获取信息——这是一种更具泛化能力的知识推理技能。由于所有提示方法仍远未达到领域内最先进方法的水平（表1），我们认为使用更多人工编写的数据进行微调可能是释放ReAct潜力的更好途径。

[4]We suspect that this could be due to the sub-optimal greedy decoding procedure, and future work using better decoding (e.g. beam search) might help address this issue.

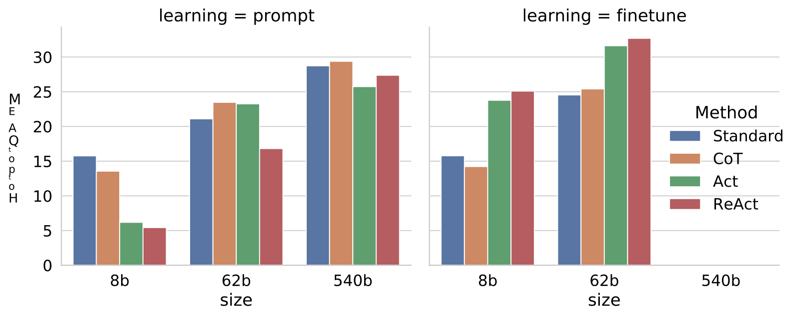

图3：在HotPotQA上使用ReAct（我们的方法）和基线进行提示和微调的扩展结果。

4 决策任务

我们还测试了ReAct在两个基于语言的交互式决策任务——ALFWorld和WebShop上的表现，这两个任务都具有复杂的环境，要求智能体在长期行动中面对稀疏奖励，因此需要推理来有效地行动和探索。

ALFWorld ALFWorld（Shridhar 等人，2020b）（图1(2)）是一个合成文本游戏，旨在与具身ALFRED基准（Shridhar 等人，2020a）对齐。它包含6种任务类型，智能体需要通过文本动作（例如，前往咖啡桌1，拿起纸张2，使用台灯1）在模拟家庭环境中导航和交互，以实现高层目标（例如，在台灯下检查纸张）。一个任务实例可能包含超过50个位置，并且需要专家策略超过50步才能解决，因此要求智能体规划并跟踪子目标，同时系统地探索（例如，逐个检查所有桌子寻找台灯）。特别是，ALFWorld内置的一个挑战是需要确定常见家居物品的可能位置（例如，台灯通常可能在桌子、架子或梳妆台上），这使得该环境非常适合利用LLM预训练的常识知识。为了提示ReAct，我们为每种任务类型从训练集中随机标注三条轨迹，每条轨迹包含稀疏的思考，这些思考（1）分解目标，（2）跟踪子目标完成情况，（3）确定下一个子目标，以及（4）通过常识推理在哪里找到物体以及如何处理它。我们在附录C.4中展示了用于ALFWorld的提示。遵循Shridhar等人（2020b）的方法，我们在任务特定的设置中评估了134个未见过的评估游戏。为了鲁棒性，我们通过从标注的三条轨迹中选取两条进行排列，为每种任务类型构建了6个提示。Act提示使用相同的轨迹构建，但不包含思考——由于任务实例是从训练集中随机选择的，这既不偏向ReAct也不偏向Act，从而提供了公平且受控的比较来测试稀疏思考的重要性。作为基线，我们使用了BUTLER（Shridhar 等人，2020b），这是一个模仿学习智能体，对每种任务类型在10^5条专家轨迹上训练[5]。

WebShop能否让ReAct在实际应用中与嘈杂的真实语言环境交互？我们研究了WebShop（Yao等人，2022），这是一个最近提出的在线购物网站环境，包含118万件真实商品和1.2万条人工指令。与ALFWorld不同，WebShop包含大量结构化和非结构化文本（例如从亚马逊抓取的商品标题、描述和选项），并要求智能体通过网页交互（例如搜索“抽屉床头柜”，选择“颜色：现代镍白”或“返回搜索”等按钮）根据用户指令（例如“我正在寻找一个带抽屉的床头柜。它应有镍饰面，价格低于140美元”）购买商品。该任务通过平均得分（所选商品覆盖所需属性的百分比在所有回合中的平均值）和成功率（所选商品满足所有要求的回合百分比）在500条测试指令上进行评估。我们设计了包含搜索、选择商品、选择选项和购买等动作的Act提示，而ReAct提示则额外包含推理步骤，以确定要探索什么、何时购买以及哪些商品选项与指令相关。示例提示见表6，附录中的模型预测见表10。我们与一种模仿学习方法（IL）进行了比较。

[5]Micheli & Fleuret (2021) finetuned a GPT-2 model on 3553 task instances and achieved a much improved performance than BUTLER, but it is trained on all task types, thus not included as a baseline.

Method	Pick	Clean	Heat	Cool	Look	Pick 2	All
Act (best of 6)	88	42	74	67	72	**41**	45
ReAct (avg)	65	39	83	76	55	24	57
ReAct (best of 6)	**92**	58	**96**	86	**78**	**41**	**71**
ReAct-IM (avg)	55	59	60	55	23	24	48
ReAct-IM (best of 6)	62	**68**	87	57	39	33	53
BUTLER$_g$ (best of 8)	33	26	70	76	17	12	22
BUTLER (best of 8)	46	39	74	**100**	22	24	37

表3：AlfWorld任务特定成功率（%）。BUTLER和BUTLER$_g$的结果来自Shridhar等人（2020b）的表4。除BUTLER使用束搜索外，所有方法均使用贪心解码。

Method	Score	SR
Act	62.3	30.1
ReAct	**66.6**	**40.0**
IL	59.9	29.1
IL+RL	62.4	28.7
Human Expert	82.1	59.6

表4：Webshop上的得分与成功率（SR）。IL/IL+RL取自Yao等人（2022）。

使用1,012条人工标注轨迹训练，以及一个模仿+强化学习（IL + RL）方法，额外使用10,587条训练指令进行训练。

结果 ReAct 在 ALFWorld（表 3）和 Webshop（表 4）上的表现均优于 Act。在 ALFWorld 上，最佳 ReAct 试验的平均成功率达到 71%，显著优于最佳 Act（45%）和 BUTLER（37%）试验。事实上，即使是表现最差的 ReAct 试验（48%）也超过了这两种方法的最佳试验。此外，ReAct 相对于 Act 的优势在六次受控试验中保持一致，相对性能提升范围从 33% 到 90%，平均为 62%。从定性角度来看，我们发现，在没有进行任何推理的情况下，Act 无法正确地将目标分解为更小的子目标，或者丢失了对当前环境状态的跟踪。比较 ReAct 和 Act 的示例轨迹见附录 D.2.1 和附录 D.2.2。

在Webshop上，一次性Act提示方法的性能已与IL和IL+RL方法相当。通过额外的稀疏推理，ReAct实现了显著更优的性能，相较于之前的最佳成功率提升了整整10%。通过检查示例，我们发现ReAct更可能通过推理来识别与指令相关的产品和选项，从而弥合嘈杂观测与行动之间的差距（例如，“对于‘节省空间的客厅储物凳’，该商品有‘39x18x18英寸’和‘蓝色’选项，似乎值得购买。”）。然而，现有方法仍远未达到专家人类的性能（表4），专家人类会进行更多的产品探索和查询重构，这对于基于提示的方法而言仍具有挑战性。

关于内部推理与外部反馈的价值 据我们所知，ReAct是首个在闭环系统中将LLM应用于交互环境的推理与行动结合的演示。最接近的前期工作可能是Huang等人（2022b）的Inner Monologue（IM），其中具身智能体的行动由其同名“内心独白”驱动。然而，IM的“内心独白”仅限于对环境状态的观察以及为实现目标智能体需完成的任务。相比之下，ReAct中用于决策的推理轨迹灵活且稀疏，能够针对不同任务诱导出多样的推理类型（见第2节）。

为了展示ReAct与IM之间的差异，并强调内部推理相对于对外部反馈的简单反应的重要性，我们使用由类似IM的密集外部反馈构成的思维模式进行了一项消融实验。如表3所示，ReAct显著优于IM式提示（ReAct-IM）（总体成功率71%对比53%），在六个任务中的五个上持续具有优势。定性观察发现，由于缺乏高层次的目标分解，ReAct-IM在判断子目标何时完成或下一个子目标应该是什么时经常出错。此外，由于缺乏常识推理，许多ReAct-IM轨迹难以确定某个物品在ALFWorld环境中可能的位置。这两种缺陷都可以在ReAct范式中得到解决。关于ReAct-IM的更多细节见附录B.2。ReAct-IM的示例提示可在附录C.4中找到，示例轨迹见附录D.2.3。

5 相关工作

用于推理的语言模型　或许最著名的利用LLM进行推理的工作是思维链（Chain-of-Thought, CoT）（Wei等人，2022），它揭示了LLM能够形成自己的“思考过程”来解决问题。此后出现了一系列后续研究，包括用于解决复杂任务的自少至多提示（least-to-most prompting）（Zhou等人，2022）、零样本CoT（zero-shot CoT）（Kojima等人，2022）以及基于自洽性的推理（reasoning with self-consistency）（Wang等人，2022a）。最近，（Madaan & Yazdanbakhsh，2022）系统研究了CoT的公式化表达与结构，并观察到符号、模式和文本的存在对CoT的有效性至关重要。其他工作也已扩展到更复杂的推理架构，超越了简单的提示方法。例如，选择-推理（Selection-Inference）（Creswell等人，2022）将推理过程分为“选择”和“推理”两个步骤。STaR（Zelikman等人，2022）通过使用模型自身生成的正确推理过程对模型进行微调，从而自举推理过程。忠实推理（Faithful reasoning）（Creswell & Shanahan，2022）将多步推理分解为三个步骤，每个步骤分别由专门的LM执行。类似的方法如Scratchpad（Nye等人，2021），通过对中间计算步骤微调LM，也在多步计算问题上展示了改进。与这些方法不同，ReAct不仅执行孤立、固定的推理，还将模型动作及其对应观察整合成一个连贯的输入流，使模型能够更准确地进行推理，并处理超越推理的任务（例如交互式决策）。

用于决策制定的语言模型 LLMs的强大能力使其能够执行超出语言生成的任务，利用LLMs作为决策制定的策略模型越来越流行，尤其是在交互式环境中。WebGPT (Nakano et al., 2021) 使用LM与网页浏览器交互、导航网页，并从ELI5 (Fan et al., 2019)中推断复杂问题的答案。与ReAct相比，WebGPT 并未明确建模思维和推理过程，而是依赖昂贵的人类反馈进行强化学习。在对话建模中，诸如BlenderBot (Shuster et al., 2022b) 和 Sparrow (Glaese et al., 2022) 这样的聊天机器人，以及像SimpleTOD (Hosseini-Asl et al., 2020) 这样的任务导向型对话系统，也训练LM在API调用方面做出决策。与ReAct不同，它们同样没有明确考虑推理过程，并且依赖昂贵的数据集和人类反馈收集来进行策略学习。相比之下，ReAct以更廉价的方式学习策略，因为决策过程仅需对推理过程进行语言描述。[6]

大语言模型在交互式和具身环境中也越来越多地被用于规划和决策。在这方面，与ReAct最相关的或许是SayCan（Ahn等人，2022）和Inner Monologue（Huang等人，2022b），它们使用LLM进行机器人动作规划和决策。在SayCan中，LLM被提示直接预测机器人可能采取的动作，然后通过基于视觉环境的可供性模型重新排序以得到最终预测。Inner Monologue通过添加同名的“内心独白”（以环境注入反馈的形式实现）进行了进一步改进。据我们所知，Inner Monologue是第一个展示这种闭环系统的工作，而ReAct在此基础上构建。然而，我们认为Inner Monologue并不真正包含内心思考——这一点将在第4节中详细阐述。我们还注意到，在交互式决策过程中利用语言作为语义丰富的输入，在其他设置下也被证明是成功的（Abramson等人，2020；Karamcheti等人，2021；Huang等人，2022a；Li等人，2022）。越来越明显的是，借助LLM，语言作为一种基本认知机制将在交互和决策中发挥关键作用。此外，LLM的进展也激发了像Reed等人（2022）那样的多功能通用智能体的开发。

6 结论

我们提出了ReAct——一种简单而有效的方法，用于在大型语言模型中协同推理和行动。通过对多跳问答、事实核查和交互式决策任务的一系列多样化实验，我们展示了ReAct带来了具有可解释决策轨迹的优越性能。尽管我们的方法简单，但具有大动作空间的复杂任务需要更多的演示才能学习好，不幸的是，这很容易超出上下文学习的输入长度限制。我们探索了在HotpotQA上的微调方法。

[6] 人类反馈　k can also be incorporated in a complementary manner but we leave 它用于未来的工作。

初步结果令人鼓舞，但学习更多高质量的人工标注将是进一步提升性能的迫切需求。通过多任务训练扩展ReAct，并将其与强化学习等互补范式结合，可以产生更强大的智能体，进一步释放LLMs在更多应用中的潜力。

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可重复性声明

我们的主要实验是在PaLM（Chowdhery等人，2022）上进行的，该模型目前并非公开可访问。为提高可复现性，我们在附录C中包含了所有使用的提示词，在附录A.1中提供了使用GPT-3（Brown等人，2020）的附加实验，以及相关的GPT-3 ReAct提示代码，地址为https://anonymous.4open.science/r/ReAct-2268/。

伦理声明

ReAct 提示大型语言模型生成比以往方法更具人类可解释性、可诊断性和可控性的任务解决轨迹。然而，将大型语言模型与动作空间连接以与外部环境（例如网络、物理环境）交互存在潜在危险，例如查询不当或私人信息，或在环境中采取有害行动。我们的实验通过将交互限制在无私人信息的特定网站（维基百科或WebShop）上，并在动作空间设计中避免任何危险动作（即模型无法在WebShop研究基准上实际购买产品，或编辑维基百科），从而最小化此类风险。我们相信，研究人员在未来设计更广泛的实验之前，应意识到这些风险。

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迈克尔·安、安东尼·布罗汉、诺亚·布朗、叶夫根·切博塔尔、奥马尔·科尔特斯、拜伦·大卫、切尔西·芬恩、楚媛·傅、基尔塔娜·戈帕拉克里希南、卡罗尔·豪斯曼、亚历克斯·赫尔佐格、丹尼尔·何、茉莉·徐、朱利安·伊巴兹、布莱恩·伊克特、亚历克斯·厄潘、埃里克·张、罗萨里奥·豪雷吉·鲁阿诺、凯尔·杰弗里、莎莉·杰斯蒙斯、尼基尔·J乔希、瑞安·朱利安、德米特里·卡拉什尼科夫、匡玉恒、李光辉、谢尔盖·莱文、陆瑶、琳达·卢、卡罗琳娜·帕拉达、彼得·帕斯特、约内尔·基亚姆鲍、卡尼什卡·拉奥、雅雷克·雷廷豪斯、迭戈·雷耶斯、皮埃尔·塞马内、尼古拉斯·西弗斯、克莱顿·谭、亚历山大·托舍夫、文森特·范霍克、夏斐、肖泰德、徐鹏、徐思纯、颜梦媛、曾安迪。《做我能做的，而非我说的话：在机器人可操作性中奠定语言基础》，2022年。网址 https://arxiv.org/abs/2204.01691。

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A 附加结果

A.1 GPT-3 实验

	PaLM-540B	GPT-3
HotpotQA (exact match)	29.4	**30.8**
ALFWorld (success rate %)	70.9	**78.4**

表5：使用PaLM-540B与GPT-3（text-davinci-002，贪婪解码）的ReAct提示结果。在HotpotQA上，我们随机抽取了500个验证问题的子集。在ALFWorld上，我们使用了全部134个未见过的验证任务实例，并根据PaLM-540B使用了最佳的提示集。

我们进行了额外的GPT-3（Brown等人，2020）实验，以确认ReAct提示的性能在不同大型语言模型上具有通用性。如表5所示，GPT-3（text-davinci-002，贪婪解码）在HotpotQA和ALFWorld上持续优于PaLM-540B，这可能是由于其经过人类指令遵循微调。这表明ReAct提示在不同任务的不同大型语言模型上均有效。这些实验的代码见https://react-lm.github.io/。

A.2 REAC T 获取关于 HOTPOTQA 的最新知识

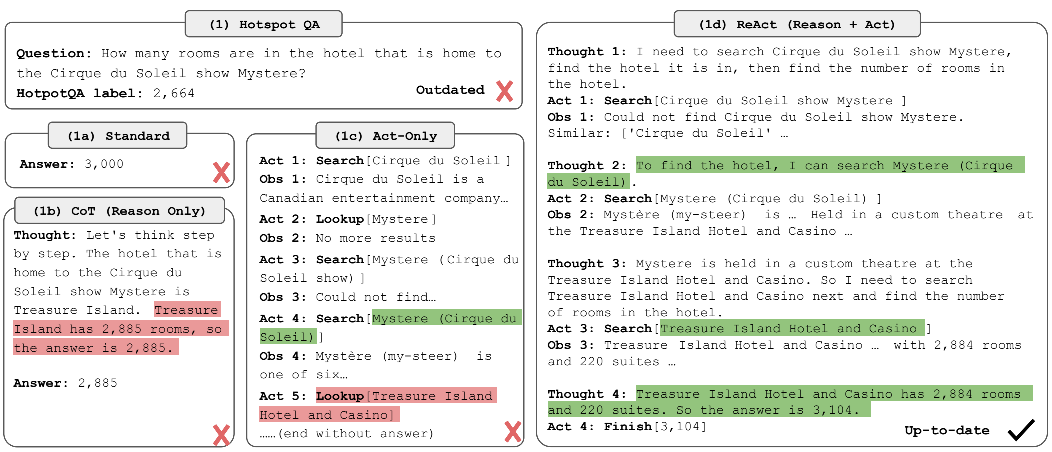

图4：另一个HotpotQA问题的示例，其中原始标签已过时。只有ReAct能够通过真实世界的网络交互和推理获得最新的答案。

在轨迹检查过程中，我们还发现有时ReAct与数据集标签不一致，因为标签本身可能已经过时。例如，如图4所示，问题询问一家酒店的面积，而该面积在HotpotQA构建后有所增加。尽管Standard和CoT因幻觉给出错误答案，Act虽然能够进行真实网络交互但仍失败，原因是缺乏推理来指导如何与互联网交互以进行问答。只有ReAct能够从互联网检索最新信息并提供合理答案。因此，更好地融入推理能力可能有助于近期互联网增强语言模型（Nakano等人，2021；Lazaridou等人，2022；Shuster等人，2022a）进行时效性任务求解。

A.3 ALFWORLD上的人类在环行为修正

我们还探索了人在回路交互与ReAct的结合，以允许人类检查和编辑ReAct的推理轨迹。图5显示，只需移除第17步中的幻觉语句，并在第23步中添加一些提示，就能使ReAct的行为发生显著变化，从而与这些人类思维编辑保持一致，并成功完成任务。从人类的角度来看，解决此类任务变得明显更加容易——从输入数十个动作变为仅编辑几个想法，这开启了人机协作的新形式。我们注意到，这种即时策略编辑

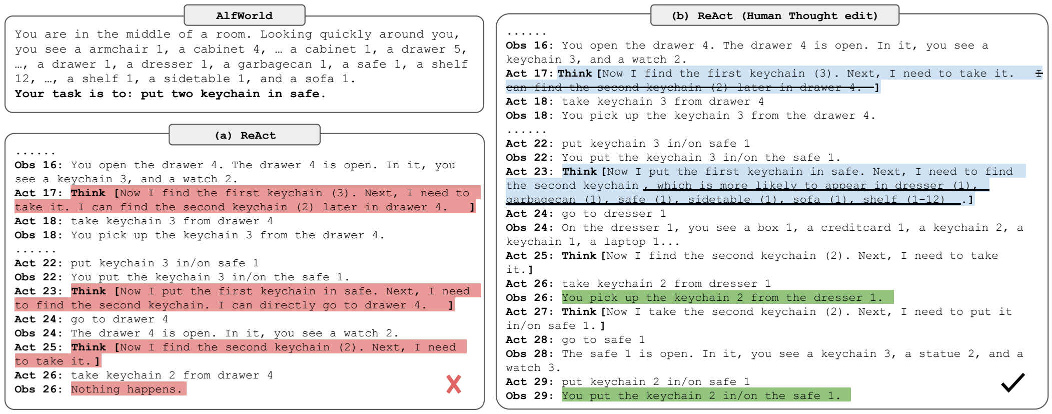

图5：在AlfWorld中使用ReAct进行人在回路行为校正示例。(a) ReAct轨迹因幻觉思想（行动17）而失败。(b) 通过人类简单编辑两个思想（行动17、23），ReAct轨迹产生理想的推理痕迹和行动并成功。

对于Act和之前的强化学习方法来说，这是困难的，因为人类无法改变模型参数，而改变少数几个动作可能无法编辑模型其余的行为。这种范式也超越了Huang等人（2022b）中通过人类对话更新目标或子目标的方式——虽然编辑ReAct的思考可以做到这些，但它还能修改模型的内部信念、推理风格，或者灵活思考空间支持的任何内容，以便更好地完成任务求解。我们认为这是实现人类对齐的一个令人兴奋的方向，并将更系统的研究留作未来工作。

B 实验细节

B.1 HOTPOTQA 微调细节

对于所有微调，我们使用批次大小为64。在PaLM-8B上，我们对ReAct和Act方法微调4,000步，对Standard和CoT方法微调2,000步。在PaLM-62B上，我们对ReAct和Act方法微调4,000步，对Standard和CoT方法微调1,000步。我们发现ReAct和Act方法通常受益于更多的训练步数（以及更多的训练数据），而Standard和CoT方法在微调后很快退化。

B.2 ALFWORLD IM-风格详情

对于IM风格的消融实验，ReAct中使用的相同专家轨迹被重新标注了密集的外部反馈思考，这些思考限制了ReAct-IM仅能够考虑：(1) 分解当前目标；(2) 当前需要完成的子目标。具体而言，ReAct-IM缺乏以下思考：(1) 判断子目标何时完成；(2) 确定下一个子目标应该是什么；(3) 引导LLM调用其内部预训练知识来识别环境中物品可能所处的位置。

C 提示

C.1 HOTPOTQA

Hotpot QA Prompts

Original	
Question	What is the elevation range for the area that the eastern sector of the Colorado orogeny extends into?
Answer	1,800 to 7,000 ft
Question	Musician and satirist Allie Goertz wrote a song about the "The Simpsons" character Milhouse, who Matt Groening named after who?
Answer	Richard Nixon
Question	Which documentary is about Finnish rock groups, Adam Clayton Powell or The Saimaa Gesture?
Answer	The Saimaa Gesture
Question	What profession does Nicholas Ray and Elia Kazan have in common?
Answer	director, screenwriter, actor
Question	Which magazine was started first Arthur's Magazine or First for Women?
Answer	Arthur's Magazine
Question	Were Pavel Urysohn and Leonid Levin known for the same type of work?
Answer	Yes
Act	
Question	What is the elevation range for the area that the eastern sector of the Colorado orogeny extends into?
Action 1	Search[Colorado orogeny]
Observation 1	The Colorado orogeny was an episode of mountain building (an orogeny) in Colorado and surrounding areas.
Action 2	Lookup[eastern sector]
Observation 2	(Result 1 / 1) The eastern sector extends into the High Plains and is called the Central Plains orogeny.
Action 3	Search[High Plains]
Observation 3	High Plains refers to one of two distinct land regions:
Action 4	Search[High Plains (United States)]
Observation 4	The High Plains are a subregion of the Great Plains. From east to west, the High Plains rise in elevation from around 1,800 to 7,000 ft (550 to 2,130 m).[3]
Action 5	Finish[1,800 to 7,000 ft]
Question	Musician and satirist Allie Goertz wrote a song about the "The Simpsons" character Milhouse, who Matt Groening named after who?
Action 1	Search[Milhouse]
Observation 1	Milhouse Mussolini Van Houten is a recurring character in the Fox animated television series The Simpsons voiced by Pamela Hayden and created by Matt Groening.
Action 2	Lookup[named after]
Observation 2	(Result 1 / 1) Milhouse was named after U.S. president Richard Nixon, whose middle name was Milhous.
Action 3	Finish[Richard Nixon]
Question	Which documentary is about Finnish rock groups, Adam Clayton Powell or The Saimaa Gesture?
Action 1	Search[Adam Clayton Powell]
Observation 1	Could not find [Adam Clayton Powell]. Similar: ['Adam Clayton Powell III', 'Seventh Avenue (Manhattan)', 'Adam Clayton Powell Jr. State Office Building', 'Isabel Washington Powell', 'Adam Powell', 'Adam Clayton Powell (film)', 'Giancarlo Esposito'].
Action 2	Search[Adam Clayton Powell (film)]
Observation 2	Adam Clayton Powell is a 1989 American documentary film directed by Richard Kilberg. The film is about the rise and fall of influential African-American politician Adam Clayton Powell Jr.[3][4] It was later aired as part of the PBS series The American Experience.
Action 3	Finish[The Saimaa Gesture]
Question	What profession does Nicholas Ray and Elia Kazan have in common?
Action 1	Search[Nicholas Ray]
Observation 1	Nicholas Ray (born Raymond Nicholas Kienzle Jr., August 7, 1911 - June 16, 1979) was an American film director, screenwriter, and actor best known for the 1955 film Rebel Without a Cause.
Action 2	Search[Elia Kazan]
Observation 2	Elia Kazan was an American film and theatre director, producer, screenwriter and actor.

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Hotpot QA Prompts – Continued from previous page

Action 3	Finish[director, screenwriter, actor]
Question	Which magazine was started first Arthur’s Magazine or First for Women?
Action 1	Search[Arthur’s Magazine]
Observation 1	Arthur’s Magazine (1844–1846) was an American literary periodical published in Philadelphia in the 19th century.
Action 2	Search[First for Women]
Observation 2	First for Women is a woman’s magazine published by Bauer Media Group in the USA.[1] The magazine was started in 1989.
Action 3	Finish[Arthur’s Magazine]
Question	Were Pavel Urysohn and Leonid Levin known for the same type of work?
Action 1	Search[Pavel Urysohn]
Observation 1	Pavel Samuilovich Urysohn (February 3, 1898 - August 17, 1924) was a Soviet mathematician who is best known for his contributions in dimension theory.
Action 2	Search[Leonid Levin]
Observation 2	Leonid Anatolievich Levin is a Soviet-American mathematician and computer scientist.
Action 3	Finish[yes]
CoT	
Question	What is the elevation range for the area that the eastern sector of the Colorado orogeny extends into?
Thought	Let’s think step by step. The eastern sector of Colorado orogeny extends into the High Plains. High Plains rise in elevation from around 1,800 to 7,000 ft, so the answer is 1,800 to 7,000 ft.
Answer	1,800 to 7,000 ft
Question	Musician and satirist Allie Goertz wrote a song about the "The Simpsons" character Milhouse, who Matt Groening named after who?
Thought	Let’s think step by step. Milhouse was named after U.S. president Richard Nixon, so the answer is Richard Nixon.
Answer	Richard Nixon
Question	Which documentary is about Finnish rock groups, Adam Clayton Powell or The Saimaa Gesture?
Thought	Let’s think step by step. Adam Clayton Powell (film) is a documentary about an African-American politician, not Finnish rock groups. So the documentary about Finnish rock groups must instead be The Saimaa Gesture.
Answer	The Saimaa Gesture
Question	What profession does Nicholas Ray and Elia Kazan have in common?
Thought	Let’s think step by step. Professions of Nicholas Ray are director, screenwriter, and actor. Professions of Elia Kazan are director, producer, screenwriter, and actor. So profession Nicholas Ray and Elia Kazan have in common is director, screenwriter, and actor.
Answer	director, screenwriter, actor
Question	Which magazine was started first Arthur’s Magazine or First for Women?
Thought	Let’s think step by step. Arthur’s Magazine was started in 1844. First for Women was started in 1989. 1844 (Arthur’s Magazine) < 1989 (First for Women), so Arthur’s Magazine was started first.
Answer	Arthur’s Magazine
Question	Were Pavel Urysohn and Leonid Levin known for the same type of work?
Thought	Let’s think step by step. Pavel Urysohn is a mathematician. Leonid Levin is a mathematician and computer scientist. So Pavel Urysohn and Leonid Levin have the same type of work.
Answer	Yes
ReAct	
Question	What is the elevation range for the area that the eastern sector of the Colorado orogeny extends into?
Thought 1	I need to search Colorado orogeny, find the area that the eastern sector of the Colorado orogeny extends into, then find the elevation range of the area.
Action 1	Search[Colorado orogeny]
Observation 1	The Colorado orogeny was an episode of mountain building (an orogeny) in Colorado and surrounding areas.
Thought 2	It does not mention the eastern sector. So I need to look up eastern sector.
Action 2	Lookup[eastern sector]
Observation 2	(Result 1 / 1) The eastern sector extends into the High Plains and is called the Central Plains orogeny.
Thought 3	The eastern sector of Colorado orogeny extends into the High Plains. So I need to search High Plains and find its elevation range.
Action 3	Search[High Plains]
Observation 3	High Plains refers to one of two distinct land regions
Thought 4	I need to instead search High Plains (United States).
Action 4	Search[High Plains (United States)]

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Hotpot QA Prompts – Continued from previous page

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Observation 4   The High Plains are a subregion of the Great Plains.  From east to west, the
                High Plains rise in elevation from around 1,800 to 7,000 ft (550 to 2,130
                m).[3]
Thought 5       High Plains rise in elevation from around 1,800 to 7,000 ft, so the answer
                is 1,800 to 7,000 ft.
Action 5        Finish[1,800 to 7,000 ft]

Question        Musician and satirist Allie Goertz wrote a song about the "The Simpsons"
                character Milhouse, who Matt Groening named after who?
Thought 1       The question simplifies to "The Simpsons" character Milhouse is named after
                who.  I only need to search Milhouse and find who it is named after.
Action 1        Search[Milhouse]
Observation 1   Milhouse Mussolini Van Houten is a recurring character in the Fox animated
                television series The Simpsons voiced by Pamela Hayden and created by Matt
                Groening.
Thought 2       The paragraph does not tell who Milhouse is named after, maybe I can look up
                "named after".
Action 2        Lookup[named after]
Observation 2   (Result 1 / 1) Milhouse was named after U.S. president Richard Nixon, whose
                middle name was Milhous.
Thought 3       Milhouse was named after U.S. president Richard Nixon, so the answer is
                Richard Nixon.
Action 3        Finish[Richard Nixon]

Question        Which documentary is about Finnish rock groups, Adam Clayton Powell or The
                Saimaa Gesture?
Thought 1       I need to search Adam Clayton Powell and The Saimaa Gesture, and find which
                documentary is about Finnish rock groups.
Action 1        Search[Adam Clayton Powell]
Observation 1   Could not find [Adam Clayton Powell].  Similar:  ['Adam Clayton Powell
                III', 'Seventh Avenue (Manhattan)', 'Adam Clayton Powell Jr.  State Office
                Building', 'Isabel Washington Powell', 'Adam Powell', 'Adam Clayton Powell
                (film)', 'Giancarlo Esposito'].
Thought 2       To find the documentary, I can search Adam Clayton Powell (film).
Action 2        Search[Adam Clayton Powell (film)]
Observation 2   Adam Clayton Powell is a 1989 American documentary film directed by
                Richard Kilberg.  The film is about the rise and fall of influential
                African-American politician Adam Clayton Powell Jr.[3][4] It was later aired
                as part of the PBS series The American Experience.
Thought 3       Adam Clayton Powell (film) is a documentary about an African-American
                politician, not Finnish rock groups.  So the documentary about Finnish rock
                groups must instead be The Saimaa Gesture.
Action 3        Finish[The Saimaa Gesture]

Question        What profession does Nicholas Ray and Elia Kazan have in common?
Thought 1       I need to search Nicholas Ray and Elia Kazan, find their professions, then
                find the profession they have in common.
Action 1        Search[Nicholas Ray]
Observation 1   Nicholas Ray (born Raymond Nicholas Kienzle Jr., August 7, 1911 - June 16,
                1979) was an American film director, screenwriter, and actor best known for
                the 1955 film Rebel Without a Cause.
Thought 2       Professions of Nicholas Ray are director, screenwriter, and actor.  I need
                to search Elia Kazan next and find his professions.
Action 2        Search[Elia Kazan]
Observation 2   Elia Kazan was an American film and theatre director, producer, screenwriter
                and actor.
Thought 3       Professions of Elia Kazan are director, producer, screenwriter, and actor.
                So profession Nicholas Ray and Elia Kazan have in common is director,
                screenwriter, and actor.
Action 3        Finish[director, screenwriter, actor]

Question        Which magazine was started first Arthur's Magazine or First for Women?
Thought 1       I need to search Arthur's Magazine and First for Women, and find which was
                started first.
Action 1        Search[Arthur's Magazine]
Observation 1   Arthur's Magazine (1844-1846) was an American literary periodical published
                in Philadelphia in the 19th century.
Thought 2       Arthur's Magazine was started in 1844.  I need to search First for Women
                next.
Action 2        Search[First for Women]
Observation 2   First for Women is a woman's magazine published by Bauer Media Group in the
                USA.[1] The magazine was started in 1989.
Thought 3       First for Women was started in 1989.  1844 (Arthur's Magazine) < 1989 (First
                for Women), so Arthur's Magazine was started first.
Action 3        Finish[Arthur's Magazine]

Question        Were Pavel Urysohn and Leonid Levin known for the same type of work?
Thought 1       I need to search Pavel Urysohn and Leonid Levin, find their types of work,
                then find if they are the same.
Action 1        Search[Pavel Urysohn]
```

Continued on next page

Hotpot QA 提示词 – 接上一页

Observation 1	Pavel Samuilovich Urysohn (February 3, 1898 – August 17, 1924) was a Soviet mathematician who is best known for his contributions in dimension theory.
Thought 2	Pavel Urysohn is a mathematician. I need to search Leonid Levin next and find its type of work.
Action 2	Search[Leonid Levin]
Observation 2	Leonid Anatolievich Levin is a Soviet-American mathematician and computer scientist.
Thought 3	Leonid Levin is a mathematician and computer scientist. So Pavel Urysohn and Leonid Levin have the same type of work.
Action 3	Finish[yes]

C.2 发烧

FEVER Prompts

Original	Determine if there is Observation that SUPPORTS or REFUTES a Claim, or if there is NOT ENOUGH INFORMATION.
Claim	Nikolaj Coster-Waldau worked with the Fox Broadcasting Company.
Answer	SUPPORTS
Claim	Stranger Things is set in Bloomington, Indiana.
Answer	REFUTES
Claim	Beautiful reached number two on the Billboard Hot 100 in 2003.
Answer	NOT ENOUGH INFO
Act	Determine if there is Observation that SUPPORTS or REFUTES a Claim, or if there is NOT ENOUGH INFORMATION.
Claim	Nikolaj Coster-Waldau worked with the Fox Broadcasting Company.
Action2	Search[Nikolaj Coster-Waldau]
Observation 1	Nikolaj William Coster-Waldau (born 27 July 1970) is a Danish actor and producer. He graduated from the Danish National School of Performing Arts in Copenhagen in 1993,[1] and had his breakthrough role in Denmark with the film Nightwatch (1994). He played Jaime Lannister in the HBO fantasy drama series Game of Thrones, for which he received two Primetime Emmy Award nominations for Outstanding Supporting Actor in a Drama Series.. Coster-Waldau has appeared in numerous films in his native Denmark and Scandinavia, including Headhunters (2011) and A Thousand Times Good Night (2013). In the U.S, his debut film role was in the war film Black Hawk Down (2001), playing Medal of Honor recipient Gary Gordon.[2] He then played a detective in the short-lived Fox television series New Amsterdam (2008), and appeared in the 2009 Fox television film Virtuality, originally intended as a pilot.
Action 2	Finish[SUPPORTS]
Claim	Stranger Things is set in Bloomington, Indiana.
Action 1	Search[Stranger Things]
Observation 1	Stranger Things is an American science fiction horror drama television series created by the Duffer Brothers. Set in the 1980s, primarily in the fictional town of Hawkins, Indiana, the series centers on a number of mysteries and supernatural events occurring around the town and their impact on an ensemble of child and adult characters.
Action 2	Finish[REFUTES]
Claim	Beautiful reached number two on the Billboard Hot 100 in 2003.
Action 1	Search[Beautiful]
Observation 1	Could not find [Beautiful]. Similar: ['Beautiful', 'Beautiful, Beautiful', 'A Beautiful Mind (film)', 'Beautiful (Christina Aguilera song)', 'Life Is Beautiful'].
Action 2	Search[Beautiful (Christina Aguilera song)
Observation 2	"Beautiful" is a song recorded by American singer Christina Aguilera for her fourth studio album, Stripped (2002).
Action 3	Lookup[Billboard Hot 100]
Observation 3	(Result 1 / 3) The song peaked at number two on the Billboard Hot 100 in the United States, where it was certified Gold for 500,000 units shipped.
Action 4	Finish[NOT ENOUGH INFO]
CoT	Determine if there is Observation that SUPPORTS or REFUTES a Claim, or if there is NOT ENOUGH INFORMATION.
Claim	Nikolaj Coster-Waldau worked with the Fox Broadcasting Company.
Thought	Nikolaj William Coster-Waldau appeared in the 2009 Fox television film Virtuality, so he has worked with the Fox Broadcasting Company.
Answer	SUPPORTS
Claim	Stranger Things is set in Bloomington, Indiana.
Thought	Stranger Things is in the fictional town of Hawkins, Indiana, not in Bloomington, Indiana.
Answer	REFUTES

接下一页

Fever Prompts – Continued from previous page

Claim	Beautiful reached number two on the Billboard Hot 100 in 2003.
Thought	The song peaked at number two on the Billboard Hot 100 in the United States, but not sure if it was in 2003.
Answer	NOT ENOUGH INFO
ReAct	Determine if there is Observation that SUPPORTS or REFUTES a Claim, or if there is NOT ENOUGH INFORMATION.
Claim	Nikolaj Coster-Waldau worked with the Fox Broadcasting Company.
Thought 1	I need to search Nikolaj Coster-Waldau and find if he has worked with the Fox Broadcasting Company.
Action 1	Search[Nikolaj Coster-Waldau]
Observation 1	Nikolaj William Coster-Waldau (born 27 July 1970) is a Danish actor and producer. He graduated from the Danish National School of Performing Arts in Copenhagen in 1993,[1] and had his breakthrough role in Denmark with the film Nightwatch (1994). He played Jaime Lannister in the HBO fantasy drama series Game of Thrones, for which he received two Primetime Emmy Award nominations for Outstanding Supporting Actor in a Drama Series.. Coster-Waldau has appeared in numerous films in his native Denmark and Scandinavia, including Headhunters (2011) and A Thousand Times Good Night (2013). In the U.S, his debut film role was in the war film Black Hawk Down (2001), playing Medal of Honor recipient Gary Gordon.[2] He then played a detective in the short-lived Fox television series New Amsterdam (2008), and appeared in the 2009 Fox television film Virtuality, originally intended as a pilot.
Thought 2	Because he "appeared in the 2009 Fox television film Virtuality", he should have worked with the Fox Broadcasting Company.
Action 2	Finish[SUPPORTS]
Claim	Stranger Things is set in Bloomington, Indiana.
Thought 1	I should search for Stranger Things, and see if it is set in Bloomington, Indiana.
Action 1	Search[Stranger Things]
Observation 1	Stranger Things is an American science fiction horror drama television series created by the Duffer Brothers. Set in the 1980s, primarily in the fictional town of Hawkins, Indiana, the series centers on a number of mysteries and supernatural events occurring around the town and their impact on an ensemble of child and adult characters.
Thought 2	The observation says that it is set in a "fictional town of Hawkins, Indiana", so it is not set in Bloomington.
Action 2	Finish[REFUTES]
Claim	Beautiful reached number two on the Billboard Hot 100 in 2003.
Thought 1	I need to search the song Beautiful and find if it reached number two on the Billboard Hot 100 in 2003.
Action 1	Search[Beautiful]
Observation 1	Could not find [Beautiful]. Similar: ['Beautiful', 'Beautiful, Beautiful', 'A Beautiful Mind (film)', 'Beautiful (Christina Aguilera song)', 'Life Is Beautiful'].
Thought 2	From suggestions, I should search "Beautiful (Christina Aguilera song)" to find the song.
Action 2	Search[Beautiful (Christina Aguilera song)]
Observation 2	"Beautiful" is a song recorded by American singer Christina Aguilera for her fourth studio album, Stripped (2002).
Thought 3	It does not mention Billboard, so I need to look up "Billboard Hot 100" to find if it reached number two on it in 2003.
Action 3	Lookup[Billboard Hot 100]
Observation 3	(Result 1 / 3) The song peaked at number two on the Billboard Hot 100 in the United States, where it was certified Gold for 500,000 units shipped.
Thought 4	It only says the song peaked at number two on the Billboard Hot 100, but not if it was in 2003. I am not sure if this claim is true or not.
Action 4	Finish[NOT ENOUGH INFO]

C.3 WEBSHOP

表6：第4节中Webshop实验使用的提示。我们在Act中使用左侧的提示，在ReAct中使用右侧的提示（添加推理）。

Instruction: i would like a 3 ounce bottle of bright citrus deodorant for sensitive skin, and price lower than 50.00 dollars

`Act`

Action: **search**[3 ounce bright citrus deodorant sensitive skin]
Observation:
[Back to Search]
Page 1 (Total results: 50)
[Next]

[B078GWRC1J]
Bright Citrus Deodorant by Earth Mama | Natural and Safe for Sensitive Skin, Pregnancy and Breastfeeding, Contains Organic Calendula 3-Ounce
$10.99
[B078GTKVXY]
Ginger Fresh Deodorant by Earth Mama | Natural and Safe for Sensitive Skin, Pregnancy and Breastfeeding, Contains Organic Calendula 3-Ounce
$10.99
[B08KBVJ4XN]
Barrel and Oak - Aluminum-Free Deodorant, Deodorant for Men , Essential Oil-Based Scent, 24-Hour Odor Protection, Cedar & Patchouli Blend, Gentle on Sensitive Skin (Mountain Sage, 2.7 oz, 2-Pack)
$15.95

Action: **click**[B078GWRC1J]
Observation:
[Back to Search]
[Prev]
scent [assorted scents][bright citrus][calming lavender][ginger [fresh] simply non-scents]
size [travel set (4-pack)][3 ounce (pack of 1)][3-ounce (2 -pack)]
Bright Citrus Deodorant by Earth Mama | Natural and Safe for Sensitive Skin, Pregnancy and Breastfeeding, Contains Organic Calendula 3-Ounce
Price: $10.99
Rating: N.A.
[Description]
[Features]
[Reviews]
[Buy Now]

Action: **click**[bright citrus]
Observation: You have clicked bright citrus.

Action: **click**[3 ounce (pack of 1)]
Observation: You have clicked 3 ounce (pack of 1).

Action: **click**[Buy Now]

`ReAct`

Action: **search**[3 ounce bright citrus deodorant sensitive skin
Observation:
[Back to Search]
Page 1 (Total results: 50)
[Next]

[B078GWRC1J]
Bright Citrus Deodorant by Earth Mama | Natural and Safe for Sensitive Skin, Pregnancy and Breastfeeding, Contains Organic Calendula 3-Ounce
$10.99
[B078GTKVXY]
Ginger Fresh Deodorant by Earth Mama | Natural and Safe for Sensitive Skin, Pregnancy and Breastfeeding, Contains Organic Calendula 3-Ounce
$10.99
[B08KBVJ4XN]
Barrel and Oak - Aluminum-Free Deodorant, Deodorant for Men , Essential Oil-Based Scent, 24-Hour Odor Protection, Cedar & Patchouli Blend, Gentle on Sensitive Skin (Mountain Sage, 2.7 oz, 2-Pack)
$15.95

Action: **think**[B078GWRC1J and B078GTKVXY are bright citrus deodorant less then 50 dollars. I can check B078GWRC1J first.
Observation: OK.

Action: **click**[B078GWRC1J]
Observation:
[Back to Search]
[Prev]
scent [assorted scents][bright citrus][calming lavender][ginger [fresh] simply non-scents]
size [travel set (4-pack)][3 ounce (pack of 1)][3-ounce (2 -pack)]
Bright Citrus Deodorant by Earth Mama | Natural and Safe for Sensitive Skin, Pregnancy and Breastfeeding, Contains Organic Calendula 3-Ounce
Price: $10.99
Rating: N.A.
[Description]
[Features]
[Reviews]
[Buy Now]

Action: **think**[For 3 ounce bottle of bright citrus deodorant for sensitive skin, the item has options 'bright citrus' and '3 ounce (pack of 1)' and seems good to buy.]
Observation: OK.

Action: **click**[bright citrus]
Observation: You have clicked bright citrus.

Action: **click**[3 ounce (pack of 1)]
Observation: You have clicked 3 ounce (pack of 1).

Action: **click**[Buy Now]

C.4 ALFWORLD

表7：在ALFWorld清洁任务上的一个Act提示。未提供任何想法。

```
You are in the middle of a room. Looking quickly around you, you see a cabinet
13, a cabinet 12, a cabinet 11, a cabinet 10, a cabinet 9, a cabinet 8, a cabinet
7, a cabinet 6, a cabinet 5, a cabinet 4, a cabinet 3, a cabinet 2, a cabinet
1, a coffeemachine 1, a countertop 1, a diningtable 1, a drawer 1, a fridge 1, a
garbagecan 1, a microwave 1, a shelf 3, a shelf 2, a shelf 1, a sinkbasin 1, a
stoveburner 4, a stoveburner 3, a stoveburner 2, a stoveburner 1, and a toaster 1.
Your task is to: put a clean lettuce in diningtable.
> go to fridge 1
The fridge 1 is closed.
> open fridge 1
You open the fridge 1. The fridge 1 is open. In it, you see a cup 3, a egg 2, a
potato 3, and a potato 2.
> go to diningtable 1
On the diningtable 1, you see a apple 1, a bread 1, a butterknife 2, a cup 2, a
fork 2, a knife 2, a knife 1, a ladle 1, a lettuce 1, a mug 2, a mug 1, a pan 2, a
peppershaker 1, a spatula 3, a tomato 2, and a tomato 1.
> take lettuce 1 from diningtable 1
You pick up the lettuce 1 from the diningtable 1.
> go to sinkbasin 1
On the sinkbasin 1, you see a apple 2, a ladle 2, a spoon 1, and a tomato 3.
> clean lettuce 1 with sinkbasin 1
You clean the lettuce 1 using the sinkbasin 1.
> think: Now I clean a lettuce (1). Next, I need to put it in/on diningtable 1.
OK.
> go to diningtable 1
On the diningtable 1, you see a apple 1, a bread 1, a butterknife 2, a cup 2, a fork
2, a knife 2, a knife 1, a ladle 1, a mug 2, a mug 1, a pan 2, a peppershaker 1, a
spatula 3, a tomato 2, and a tomato 1.
> put lettuce 1 in/on diningtable 1
You put the lettuce 1 in/on the diningtable 1.
```

表8：ALFWorld清洁任务上的ReAct提示。

```
You are in the middle of a room. Looking quickly around you, you see a cabinet
13, a cabinet 12, a cabinet 11, a cabinet 10, a cabinet 9, a cabinet 8, a cabinet
7, a cabinet 6, a cabinet 5, a cabinet 4, a cabinet 3, a cabinet 2, a cabinet
1, a coffeemachine 1, a countertop 1, a diningtable 1, a drawer 1, a fridge 1, a
garbagecan 1, a microwave 1, a shelf 3, a shelf 2, a shelf 1, a sinkbasin 1, a
stoveburner 4, a stoveburner 3, a stoveburner 2, a stoveburner 1, and a toaster 1.
Your task is to: put a clean lettuce in diningtable.
> think: To solve the task, I need to find and take a lettuce, then clean it with
sinkbasin, then put it in diningtable.
OK.
> think: First I need to find a lettuce. A lettuce is more likely to appear in
fridge (1), diningtable (1), sinkbasin (1), stoveburner (1-3), cabinet (1-13). I
can check one by one, starting with fridge 1.
OK.
> go to fridge 1
The fridge 1 is closed.
> open fridge 1
You open the fridge 1. The fridge 1 is open. In it, you see a cup 3, a egg 2, a
potato 3, and a potato 2.
> go to diningtable 1
On the diningtable 1, you see a apple 1, a bread 1, a butterknife 2, a cup 2, a
fork 2, a knife 2, a knife 1, a ladle 1, a lettuce 1, a mug 2, a mug 1, a pan 2, a
peppershaker 1, a spatula 3, a tomato 2, and a tomato 1.
> think: Now I find a lettuce (1). Next, I need to take it.
OK.
> take lettuce 1 from diningtable 1
You pick up the lettuce 1 from the diningtable 1.
> think: Now I take a lettuce (1). Next, I need to go to sinkbasin (1) and clean
it.
OK.
> go to sinkbasin 1
On the sinkbasin 1, you see a apple 2, a ladle 2, a spoon 1, and a tomato 3.
> clean lettuce 1 with sinkbasin 1
You clean the lettuce 1 using the sinkbasin 1.
> think: Now I clean a lettuce (1). Next, I need to put it in/on diningtable 1.
OK.
> go to diningtable 1
On the diningtable 1, you see a apple 1, a bread 1, a butterknife 2, a cup 2, a fork
2, a knife 2, a knife 1, a ladle 1, a mug 2, a mug 1, a pan 2, a peppershaker 1, a
spatula 3, a tomato 2, and a tomato 1.
> put lettuce 1 in/on diningtable 1
You put the lettuce 1 in/on the diningtable 1.
```

表9：ALFWorld清洁任务中的ReAct-IM提示。

```
You are in the middle of a room.  Looking quickly around you, you see a cabinet
13, a cabinet 12, a cabinet 11, a cabinet 10, a cabinet 9, a cabinet 8, a cabinet
7, a cabinet 6, a cabinet 5, a cabinet 4, a cabinet 3, a cabinet 2, a cabinet
1, a coffeemachine 1, a countertop 1, a diningtable 1, a drawer 1, a fridge 1, a
garbagecan 1, a microwave 1, a shelf 3, a shelf 2, a shelf 1, a sinkbasin 1, a
stoveburner 4, a stoveburner 3, a stoveburner 2, a stoveburner 1, and a toaster 1.
Your task is to:  put a clean lettuce in diningtable.
> think:  To solve the task, I need to find and take a lettuce, then clean it with
sinkbasin, then put it in diningtable.
OK.
> think:  First I need to find a lettuce.
OK.
> go to fridge 1
The fridge 1 is closed.
> think:  First I need to find a lettuce.
OK.
> open fridge 1
You open the fridge 1.  The fridge 1 is open.  In it, you see a cup 3, a egg 2, a
potato 3, and a potato 2.
> think:  First I need to find a lettuce.
OK.
> go to diningtable 1
On the diningtable 1, you see a apple 1, a bread 1, a butterknife 2, a cup 2, a
fork 2, a knife 2, a knife 1, a ladle 1, a lettuce 1, a mug 2, a mug 1, a pan 2, a
peppershaker 1, a spatula 3, a tomato 2, and a tomato 1.
> think:  First I need to find a lettuce.
OK.
> take lettuce 1 from diningtable 1
You pick up the lettuce 1 from the diningtable 1.
> think:  I need to clean this lettuce (1) using sinkbasin 1.
OK.
> go to sinkbasin 1
On the sinkbasin 1, you see a apple 2, a ladle 2, a spoon 1, and a tomato 3.
> think:  I need to clean this lettuce (1) using sinkbasin 1.
OK.
> clean lettuce 1 with sinkbasin 1
You clean the lettuce 1 using the sinkbasin 1.
> think:  I need to put this cleaned lettuce (1) in/on diningtable 1.
OK.
> go to diningtable 1
On the diningtable 1, you see a apple 1, a bread 1, a butterknife 2, a cup 2, a fork
2, a knife 2, a knife 1, a ladle 1, a mug 2, a mug 1, a pan 2, a peppershaker 1, a
spatula 3, a tomato 2, and a tomato 1.
> think:  I need to put this cleaned lettuce (1) in/on diningtable 1.
OK.
> put lettuce 1 in/on diningtable 1
You put the lettuce 1 in/on the diningtable 1.
```

D轨迹

D.1 发热轨迹

We provide some trajectories produced by `ReAct`, `CoT`and `Act`on several randomly selected examples from the FEVER dev set below. Search results are omitted to space.

Example 2491 (gt: REFUTES)

ReAct	
Claim:	Bermuda Triangle is a loosely-defined region in the Pacific Ocean.
Thought 1:	I need to search Bermuda Triangle and find if it is in the Pacific Ocean.
Action 1:	Search[Bermuda Triangle]
Observation 1:	Reportedly haunted locations...
Thought 2:	The observation says that it is in the western part of the North Atlantic Ocean, so it is not in the Pacific Ocean.
Action 2:	Finish[REFUTES]
Observation 2:	Episode finished
Act	
Claim:	Bermuda Triangle is a loosely-defined region in the Pacific Ocean.
Action 1	: Search[Bermuda Triangle]
Observation 1	: Reportedly haunted locations:...
Action 2:	Finish[REFUTES]
Observation 2:	Episode finished
CoT	
Claim:	Bermuda Triangle is a loosely-defined region in the Pacific Ocean.
Thought:	Let's think step by step. The Bermuda Triangle is a loosely-defined region in the Atlantic Ocean. The Pacific Ocean is not the Atlantic Ocean.
Answer:	REFUTES

Example 5908 (gt: SUPPORTS)

ReAct	
Claim:	Princess Mononoke is a film.
Thought 1:	I need to search Princess Mononoke and find if it is a film.
Action 1:	Search[Princess Mononoke]
Observation 1:	Princess Mononoke ...
Thought 2:	From the observation, it says that Princess Mononoke is a film.
Action 2:	Finish[SUPPORTS]
Observation 2:	Episode finished
Act	
Claim:	Princess Mononoke is a film.
Action 1:	Search[Princess Mononoke]
Observation 1:	Princess Mononoke ...
Action 2:	Finish[SUPPORTS]
Observation 2:	Episode finished
CoT	
Claim:	Princess Mononoke is a film.
Thought:	Let's think step by step. Princess Mononoke is a movie, and movies are films.
Answer:	SUPPORTS

Example 1951 (gt: REFUTES)

ReAct	
Claim:	Soyuz was part of the American space program.
Thought 1:	I need to search Soyuz and find if it was part of the American space program.
Action 1:	Search[Soyuz]
Observation 1:	Soyuz is a transliteration...
Thought 2:	It does not mention the American space program, so I need to search "American space program" to find if it was part of it.
Action 2:	Search[American space program]
Observation 2:	The space policy...
Thought 3:	It does not mention Soyuz, so I am not sure if this claim is true or not.
Action 3:	Finish[NOT ENOUGH INFO]
Observation 3:	Episode finished
Act	
Claim:	Soyuz was part of the American space program.
Action 1:	Search[Soyuz]
Observation 1:	Soyuz is a transliteration...
Action 2:	Finish[NOT ENOUGH INFO]
Observation 2:	Episode finished
CoT	
Claim:	Soyuz was part of the American space program.
Thought:	Let's think step by step. Soyuz is a Russian spacecraft. The American space program is NASA. NASA and Russia have worked together on the International Space Station.
Answer:	SUPPORTS

Example 3208 (gt: REFUTES)

```
ReAct
Claim:            Reign Over Me is an American film made in 2010.
Thought 1:        I need to search for Reign Over Me and find if it is an American film made
                  in 2010.
Action 1:         Search[Reign Over Me]
Observation 1:    Reign Over Me ...
Thought 2:        The observation says that it is an American film made in 2007, so it is not
                  made in 2010.
Action 2:         Finish[REFUTES]
Observation 2:    Episode finished
```

```
Act
Claim:            Reign Over Me is an American film made in 2010.
Action 1:         Search[Reign Over Me]
Action 2:         Finish[SUPPORTS]
Observation 2:    Episode finished
```

```
CoT
Claim:            Reign Over Me is an American film made in 2010.
Thought:          Let's think step by step.  First, is Reign Over Me an American film?  Yes,
                  it is.  Second, was it made in 2010?  Yes, it was.
Answer:           SUPPORTS
```

D.2 ALFWORLD TRAJECTORIES

We provide an example trajectory of `ReAct`, `Act`, and `ReAct-IM` on the same ALFWorld game, `pick_clean_then_place_in_recep-Knife-None-CounterTop-10/trial_T20190909_110445_675754`.

D.2.1 REACT TRAJECTORY

ReAct 是 abl e to correctly find the knife, clean it, and then place it on a countertop。

ReAct ALFWorld 轨迹

你身处一个房间中央。快速环顾四周，你看到一个橱柜6、一个橱柜5、一个橱柜4、一个橱柜3、一个橱柜2、一个橱柜1、一个咖啡机1、一个台面3、一个台面2、一个台面1、一个抽屉3、一个抽屉2、一个抽屉1、一个冰箱1、一个垃圾桶1、一个微波炉1、一个架子3、一个架子2、一个架子1、一个水槽1、一个炉灶燃烧器4、一个炉灶燃烧器3、一个炉灶燃烧器2、一个炉灶燃烧器1，以及一个烤面包机1。你的任务是：将一把干净的刀放在台面上。>思考：要完成任务，我需要找到并拿取一把刀，然后在水槽中清洗它，最后放在台面上。好的。>思考：首先我需要找到一把刀。刀更可能出现在橱柜（1-6）、抽屉（1-3）、台面（1-3）、冰箱（1）、垃圾桶（1）、架子（1-3）、水槽（1）、炉灶燃烧器（1-4）、烤面包机（1）中。我可以逐个检查，从橱柜1开始。好的。>前往橱柜1 在橱柜1上，你看到一个碗1。>前往橱柜2 橱柜2是关闭的。>前往橱柜3 在橱柜3上，你看到一个玻璃瓶1。>前往橱柜4 在橱柜4上，你看到一个马克杯1。>前往橱柜5 橱柜5是关闭的。>前往橱柜6 橱柜6是关闭的。>前往抽屉1 抽屉1是关闭的。>前往抽屉2 抽屉2是关闭的。>前往抽屉3 抽屉3是关闭的。>前往台面1 在台面1上，你看到生菜2、一个马克杯2、一个胡椒瓶1，以及一个勺子2。>前往台面2 在台面2上，你看到一个杯子1、一个洗碗海绵1、一个玻璃瓶3、一把刀1、一个盘子2、一个土豆3，以及一个雕像1。>思考：现在我找到了一把刀（1）。接下来，我需要拿取它。好的。>从台面2拿取刀1 你从台面2上拿起了刀1。>思考：现在我拿取了一把刀（1）。接下来，我需要前往水槽（1）并清洗它。好的。>前往水槽1 在水槽1上，你看到一个叉子3、生菜3，以及一个铲子2。>用水槽1清洗刀1 你用水槽1清洗了刀1。>思考：现在我清洗了一把刀（1）。接下来，我需要把它放在台面1上。好的。>前往台面1 在台面1上，你看到生菜2、一个马克杯2、一个胡椒瓶1，以及一个勺子2。>将刀1放在台面1上 你把刀1放在了台面1上。

D.2.2 ACT TRAJECTORY

Act is able to correctly find the knife, but tries to clean it using a sinkbasin before actually going to the sinkbasin. From there on, the model is unable to recover, and is stuck repeating the same sequence of commands. With ReAct, an existing thought would have concretely identified that the knife had been taken, and that the next subgoal was to go to a sinkbasin to clean it.

Act ALFWorld trajectory

```
You are in the middle of a room.  Looking quickly around you, you see a cabinet 6, a
cabinet 5, a cabinet 4, a cabinet 3, a cabinet 2, a cabinet 1, a coffeemachine 1, a
countertop 3, a countertop 2, a countertop 1, a drawer 3, a drawer 2, a drawer 1, a
fridge 1, a garbagecan 1, a microwave 1, a shelf 3, a shelf 2, a shelf 1, a sinkbasin
1, a stoveburner 4, a stoveburner 3, a stoveburner 2, a stoveburner 1, and a toaster
1.
Your task is to:  put a clean knife in countertop.
> go to countertop 1
On the countertop 1, you see a lettuce 2, a mug 2, a peppershaker 1, and a spoon 2.
> go to countertop 2
On the countertop 2, you see a cup 1, a dishsponge 1, a glassbottle 3, a knife 1, a
plate 2, a potato 3, and a statue 1.
> take knife 1 from countertop 2
You pick up the knife 1 from the countertop 2.
> go to countertop 1
On the countertop 1, you see a lettuce 2, a mug 2, a peppershaker 1, and a spoon 2.
> clean knife 1 with sinkbasin 1
Nothing happens.
> go to countertop 3
On the countertop 3, you see a bread 3, a butterknife 2, a cellphone 1, a creditcard
1, a fork 2, a houseplant 1, a knife 2, a spatula 1, a statue 3, a tomato 3, a tomato
2, a tomato 1, and a vase 2.
> take knife 2 from countertop 3
Nothing happens.
> go to countertop 2
On the countertop 2, you see a cup 1, a dishsponge 1, a glassbottle 3, a plate 2, a
potato 3, and a statue 1.
> take knife 1 from countertop 2
Nothing happens.
> go to countertop 1
On the countertop 1, you see a lettuce 2, a mug 2, a peppershaker 1, and a spoon 2.
> clean knife 1 with sinkbasin 1
Nothing happens.
> go to countertop 3
On the countertop 3, you see a bread 3, a butterknife 2, a cellphone 1, a creditcard
1, a fork 2, a houseplant 1, a knife 2, a spatula 1, a statue 3, a tomato 3, a tomato
2, a tomato 1, and a vase 2.
> take knife 2 from countertop 3
Nothing happens.
> go to countertop 2
On the countertop 2, you see a cup 1, a dishsponge 1, a glassbottle 3, a plate 2, a
potato 3, and a statue 1.
> take knife 1 from countertop 2
Nothing happens.
> go to countertop 1
On the countertop 1, you see a lettuce 2, a mug 2, a peppershaker 1, and a spoon 2.
> clean knife 1 with sinkbasin 1
Nothing happens.
> go to countertop 3
On the countertop 3, you see a bread 3, a butterknife 2, a cellphone 1, a creditcard
1, a fork 2, a houseplant 1, a knife 2, a spatula 1, a statue 3, a tomato 3, a tomato
2, a tomato 1, and a vase 2.
> take knife 2 from countertop 3
Nothing happens.
> go to countertop 2
On the countertop 2, you see a cup 1, a dishsponge 1, a glassbottle 3, a plate 2, a
potato 3, and a statue 1.
> take knife 1 from countertop 2
Nothing happens.
> go to countertop 1
On the countertop 1, you see a lettuce 2, a mug 2, a peppershaker 1, and a spoon 2.
> clean knife 1 with sinkbasin 1
Nothing happens.
```

D.2.3 REAC T-IM 轨迹

ReAct-IM能够正确找到刀，但无法同时清洁它。然后LLM被卡住，试图将这把刀放在台面上。一个可能的问题是不正确的想法“我需要找到一把干净的刀”，这似乎让LLM误以为刀已被清洁。在ReAct中，这一系列子目标会被每个想法具体地捕捉。

ReAct-IM ALFWorld trajectory

```
You are in the middle of a room.  Looking quickly around you, you see a cabinet 6, a
cabinet 5, a cabinet 4, a cabinet 3, a cabinet 2, a cabinet 1, a coffeemachine 1, a
countertop 3, a countertop 2, a countertop 1, a drawer 3, a drawer 2, a drawer 1, a
fridge 1, a garbagecan 1, a microwave 1, a shelf 3, a shelf 2, a shelf 1, a sinkbasin
1, a stoveburner 4, a stoveburner 3, a stoveburner 2, a stoveburner 1, and a toaster
1.
Your task is to:  put a clean knife in countertop.
> think:  To solve the task, I need to find and take a clean knife, then put it in
countertop.
OK.
> think:  I need to find a clean knife.
OK.
> go to countertop 1
On the countertop 1, you see a lettuce 2, a mug 2, a peppershaker 1, and a spoon 2.
> think:  I need to find a clean knife.
OK.
> go to countertop 2
On the countertop 2, you see a cup 1, a dishsponge 1, a glassbottle 3, a knife 1, a
plate 2, a potato 3, and a statue 1.
> think:  I need to find a clean knife.
OK.
> take knife 1 from countertop 2
You pick up the knife 1 from the countertop 2.
> think:  I need to put this knife (1) in/on countertop 1.
OK.
> go to countertop 1
On the countertop 1, you see a lettuce 2, a mug 2, a peppershaker 1, and a spoon 2.
> think:  I need to put this knife (1) in/on countertop 1.
OK.
> put knife 1 in/on countertop 1
You put the knife 1 in/on the countertop 1.
> think:  I need to put this knife (1) in/on countertop 1.
OK.
> put knife 1 in/on countertop 1
Nothing happens.
> think:  I need to put this knife (1) in/on countertop 1.
OK.
> put knife 1 in/on countertop 1
Nothing happens.
> think:  I need to put this knife (1) in/on countertop 1.
OK.
> put knife 1 in/on countertop 1
Nothing happens.
> think:  I need to put this knife (1) in/on countertop 1.
OK.
> put knife 1 in/on countertop 1
Nothing happens.
> think:  I need to put this knife (1) in/on countertop 1.
OK.
> put knife 1 in/on countertop 1
Nothing happens.
> think:  I need to put this knife (1) in/on countertop 1.
OK.
```

D.3 WEBSHOP TRAJECTORIES

表10：Act和ReAct预测的Webshop示例轨迹。与Act相比，ReAct利用推理来寻找满足所有目标属性的产品。

Instruction: get me a sixteen pack of apple cinnamon freeze dried banana chips, and price lower than 50.00 dollars

Act

Action: **search**[sixteen pack apple cinnamon freeze dried banana chips]
Observation:
[Back to Search]
Page 1 (Total results: 50)
[Next]

[B0061IVFZE]
Brothers-ALL-Natural Fruit Crisps, Strawberry Banana, 0.42 - Ounce (Pack of 100)
$85.0
[B096H2P6G2]
Moon Fruit Freeze Dried Fruit Snacks. Fruit Snacks for Kids - (Variety Pack)
$18.99
[B092JLLYK6]
Nature's Turn Freeze-Dried Fruit Snacks - Banana Crisps - Perfect For School Lunches or an On-The-Go Snack - No Sugar Added, Non GMO, Gluten Free, Nothing Artificial (0.53oz) 6-Pack
$12.99

Action: **click**[B0061IVFZE]
Observation:
[Back to Search]
[Prev]
flavor name [asian pear][banana][fuji apple & cinnamon][strawberry banana] Brothers-ALL-Natural Fruit Crisps, Strawberry Banana, 0.42 Ounce (Pack of 100)
Price: $85.0
Rating: N.A.
[Description]
[Features]
[Reviews]
[Buy Now]

Action: **click**[Buy Now]

Score: 0.125

ReAct

Action: **search**[sixteen pack apple cinnamon freeze dried banana chips]
Observation:
[Back to Search]
Page 1 (Total results: 50)
[Next]

[B0061IVFZE]
Brothers-ALL-Natural Fruit Crisps, Strawberry Banana, 0.42 - Ounce (Pack of 100)
$85.0
[B096H2P6G2]
Moon Fruit Freeze Dried Fruit Snacks. Fruit Snacks for Kids - (Variety Pack)
$18.99
[B092JLLYK6]
Nature's Turn Freeze-Dried Fruit Snacks - Banana Crisps - Perfect For School Lunches or an On-The-Go Snack - No Sugar Added, Non GMO, Gluten Free, Nothing Artificial (0.53oz) 6-Pack
$12.99

Action: **think**[B0061IVFZE is strawberry banana, not apple cinnamon. B096H2P6G2 is fruit snacks, not freeze dried banana chips. B092JLLYK6 is banana crisps, not apple cinnamon. I can check B092JLLYK6 first.
Observation: OK.

Action: **click**[B092JLLYK6]
Observation:
[Back to Search]
[Prev]
flavor name [apple][apple cinnamon][banana][cantaloupe][peach][pear][strawberry][strawberry banana][sampler variety pack][mega variety pack][orchard variety pack]
size [0.53 ounce (pack of 6)][0.53 ounce (pack of 8)][0.53 ounce (pack of 16)]
Nature's Turn Freeze-Dried Fruit Snacks - Banana Crisps - Perfect For School Lunches or an On-The-Go Snack - No Sugar Added, Non GMO, Gluten Free, Nothing Artificial (0.53oz) 6-Pack
Price: $12.99
Rating: N.A.
[Description]
[Features]
[Reviews]
[Buy Now]

Action: **think**[For sixteen pack of apple cinnamon freeze dried banana chips, the item has options 'apple cinnamon' and '0.53 ounce (pack of 16)' and seems good to buy.]
Observation: OK.

Action: **click**[apple cinnamon]
Observation: You have clicked apple cinnamon.

Action: **click**[0.53 ounce (pack of 16)]
Observation: You have clicked 0.53 ounce (pack of 16).

Action: **click**[Buy Now]

Score: 1.0

E 更多分析

E.1 成功与失败模式分析

我们提供了对应于第3.3节中成功与错误模式分析的一些示例。为节省篇幅，省略了搜索结果和非代表性步骤。

Success: True positive

ReAct	
Question:	Author David Chanoff has collaborated with a U.S. Navy admiral who served as the ambassador to the United Kingdom under which President?
Thought 1:	I need to search David Chanoff and find the U.S. Navy admiral he collaborated with.
Action 1:	Search[David Chanoff]
Observation 1:	David Chanoff is a noted author...
Action 2:	Search[William J. Crowe]
Observation 2:	William J. Crowe is...
Thought 3:	William J. Crowe served as the ambassador to the United Kingdom under President Bill Clinton. So the answer is Bill Clinton.
Action 3:	Finish[Bill Clinton]
Observation 3:	Episode finished
CoT	
Question:	What was the name of the team that represented an Arizona university in the conference that covers a broad expanse of the western United States, including Arizona, California, and New Mexico?
Thought:	Let's think step by step. The name of the team that represented an Arizona university in the conference that covers a broad expanse of the western United States, including Arizona, California, and New Mexico is Arizona State Sun Devils.
Answer:	Arizona State Sun Devils

Success: False positive

ReAct	
Question:	Did the board game San Marco or About Time come out first?
Thought 1:	I need to search San Marco and About Time, and find which came out first.
Action 1:	Search[San Marco]
Observation 1:	San Marco is one of the ... The was built in venice.
Thought 2:	San Marco is a sestiere of Venice. I need to search About Time next.
Action 2:	Search[About Time]
Observation 2:	Could not find [About Time]. Similar: [...].
Thought 3:	About Time is a 2013 film. So San Marco came out first.
Action 3:	Finish[San Marco]
Observation 3:	Episode finished
CoT	
Question:	Was Vanderbilt University or Emory University founded first?
Thought:	Let's think step by step. Vanderbilt University was founded in 1873. Emory University was founded in 1836. 1873 (Vanderbilt University) > 1836 (Emory University), so Vanderbilt University was founded first.
Answer:	Emory University

Failure: Reasoning error

ReAct	
Question:	What was the middle name of the actress who was cast alongside Harvey Fierstein, Linda Hart, Dick Latessa, Clarke Thorell, Mary Bond Davis, Laura Bell Bundy, Matthew Morrison, Corey Reynolds, and Marissa Jaret Winokur in Hairspray?
Thought 1:	I need to search Harvey Fierstein, Linda Hart, Dick Latessa, Clarke Thorell, Mary Bond Davis, Laura Bell Bundy, Matthew Morrison, Corey Reynolds, and Marissa Jaret Winokur, find the actress that was cast alongside them in Hairspray, then find its middle name.
Action 1:	Search[Harvey Fierstein]
Observation 1:	Harvey Forbes Fierstein ...
Thought 2:	Harvey Fierstein is an actor, playwright, and screenwriter. I need to search Linda Hart next...
Action 2:	Search[Linda Hart]...
CoT	
Question:	Who is older Danny Green or James Worthy?
Thought:	Let's think step by step. Danny Green was born in 1987. James Worthy was born in 1961. 1987 (Danny Green) > 1961 (James Worthy), so Danny Green is older.
Answer:	Danny Green

Failure: Search error

ReAct	
Question:	What does the goddess associated with the goddess frigg consists of what tales?
Thought 1:	I need to search goddess frigg and find its associated goddess.
Action 1:	Search[goddess frigg]
Observation 1:	Could not find goddess frigg. Similar: ['Frigg', 'Frigg and Freyja common origin hypothesis', 'Freyja (section Relation to Frigg and other goddesses and figures)', 'Eir (redirect from Eir (goddess))', 'Baldr'].
Thought 2:	I need to search Frigg instead...

Failure: Hallucination

CoT	
Question:	What was the first year a scientific journal published by an organization located in the Bronx was published?
Thought:	Let's think step by step. The first year a scientific journal published by an organization located in the Bronx was published was 1916.
Answer:	1916 (Label: 1909)

Failure: Label ambiguity

ReAct	
Question:	What nationality is the man who developed the dictator game?
Thought 1:	I need to search dictator game, find the man who developed it, then find his nationality.
Action 1:	Search[dictator game]
Observation 1:	The dictator game is a popular experimental instrument in social psychology and economics...
Action 3:	Search[Amos Tversky]
Observation 3:	Amos Nathan Tversky...
Thought 4:	Amos Tversky is an Israeli cognitive and mathematical psychologist. So the man who developed the dictator game is Israeli.
Action 4:	Finish[Israeli] (label: Israel-American)
CoT	
Question:	Kurt Fuller from Stockton, CA appeared in which American Detective Comedy Drama television series created by Steve Franks?
Thought:	Let's think step by step. Kurt Fuller from Stockton, CA appeared in Psych. Psych is an American Detective Comedy Drama television series created by Steve Franks.
Answer:	Psych (label: Psych is an American detective comedy-drama)